Weiwu Yan

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Education

University of Chicago Fall 2023 – Spring 2025
- Master of Arts in Computational Social Science

University of California, San Diego Fall 2019 – Spring 2023
- Bachelor of Science in Social Psychology
- Bachelor of Arts in Sociology

Projects

Gender and Emotional Expression in Video Game Dialogue Spring 2025
- Collected and processed 247,644 dialogue lines from 3,043 characters across 18 role-playing video games from the Video Game Dialogue Corpus.
- Conducted emotion classification using RoBERTa to analyze gender differences in emotional expression.
- Performed Part-of-Speech tagging to identify the verbs most uniquely associated with each gender.
- Utilized regular expressions to examine gender differences in the distribution of terminal punctuation.

Multiclass Classification of Scholarly Articles Spring 2024
- Collected and processed metadata of 1.7 million scholarly articles from the arXiv Dataset.
- Performed Multinomial Logistic Regression on AWS to identify latent topics across articles.
- Trained a Latent Dirichlet Allocation model on AWS to classify articles into research fields.

Funding and Academic Output in Behavioral and Cognitive Sciences Winter 2023
- Collected and processed 4,959 funding records from U.S. NSF Division of Behavioral and Cognitive Sciences.
- Scraped metadata of 207,370 scholarly articles authored by 3,230 identified researchers from Google Scholar.
- Applied k-means clustering and TF-IDF vectorization to examine the relationship between 7 identified fields and funding amounts.
- Utilized a Structural Topic Model to analyze temporal trends across 12 identified topics.
- Performed document embeddings using SciBERT to assess the impact of funding on research diversity.

Multiclass Classification of Video Games Winter 2023
- Collected and processed metadata of 75,776 video games from the Steam Games Dataset.
- Trained a Random Forest Classifier to classify games based on textual descriptions.
- Trained a Convolutional Neural Network to classify games based on cover art.

Skills

Languages: Chinese • English

Programming: Python • SQL • R

Machine Learning: Hugging Face • TensorFlow • PyTorch • scikit-learn • spaCy

Data Analysis: Pandas • NumPy • Excel • SciPy • dplyr

Data Visualization: Plotly • ggplot2 • Tableau • Matplotlib